



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE
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Year / Branch / : Cse

Semester: IV

Academic Year: 2025-2026

INDEX

S.No.	Date	Title
1.		Azure Devops Environment Setup.
2.		Azure Devops Project Setup and User Story Management.
3.		Setting Up Epics, Features, And User Stories for Project Planning.
4.		Sprint Planning.
5.		Poker Estimation.
6.		Designing Class and Sequence Diagrams for Project Architecture.
7.		Designing Architectural and ER Diagrams for Project Structure.
8.		Testing – Test Plans and Test Cases.
9.		Load Testing and Pipelines.
10.		GitHub: Project Structure & Naming Conventions.

EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP
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Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

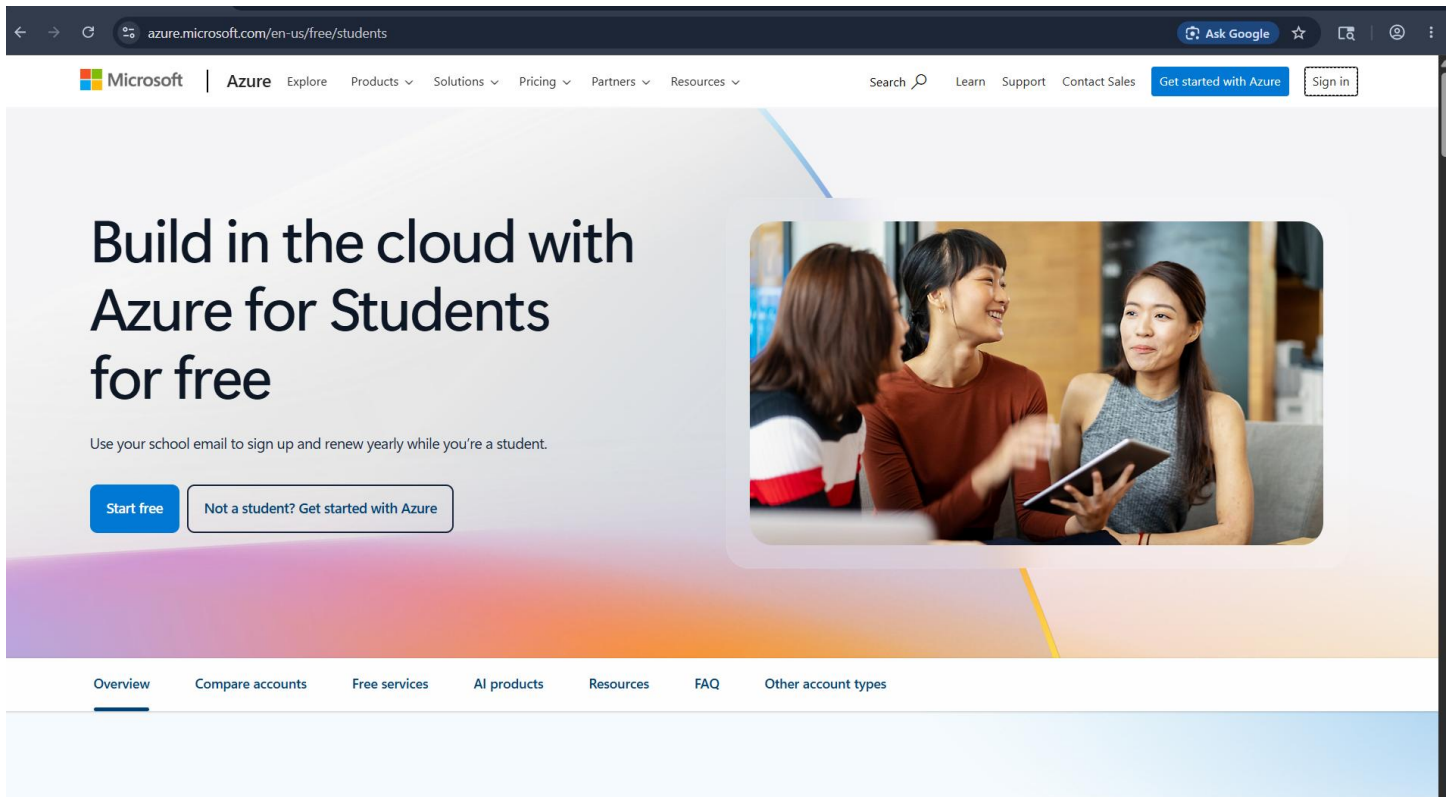
INSTALLATION

1. Open your web browser and go to the Azure website:

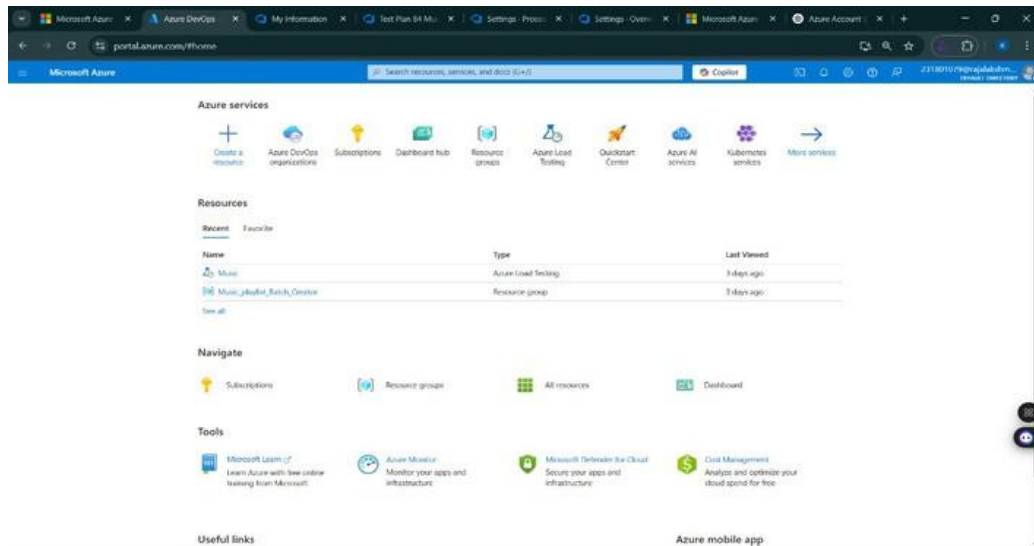
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

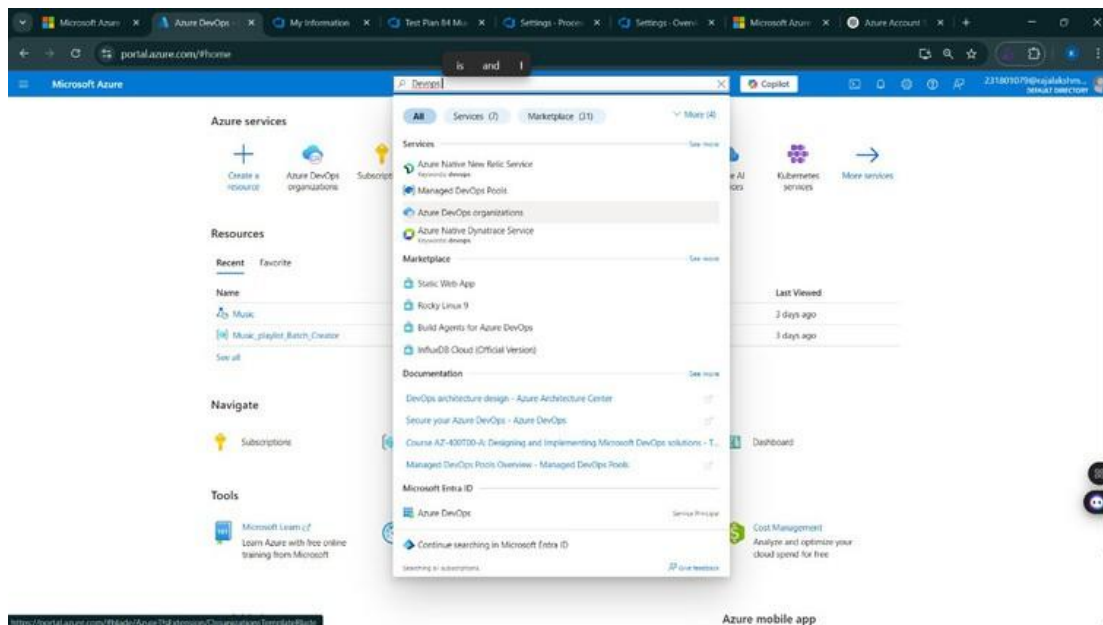
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>

A screenshot of the Azure for Students website. The browser address bar shows 'azure.microsoft.com/en-us/free/students'. The page features a large heading 'Build in the cloud with Azure for Students for free' and a subtext 'Use your school email to sign up and renew yearly while you're a student.' Below this are two buttons: 'Start free' and 'Not a student? Get started with Azure'. To the right is an image of three young women looking at a tablet. The top navigation bar includes the Microsoft logo, 'Azure', and various links like 'Explore', 'Products', 'Solutions', 'Pricing', 'Partners', and 'Resources'. The bottom navigation bar has links for 'Overview', 'Compare accounts', 'Free services', 'AI products', 'Resources', 'FAQ', and 'Other account types'.

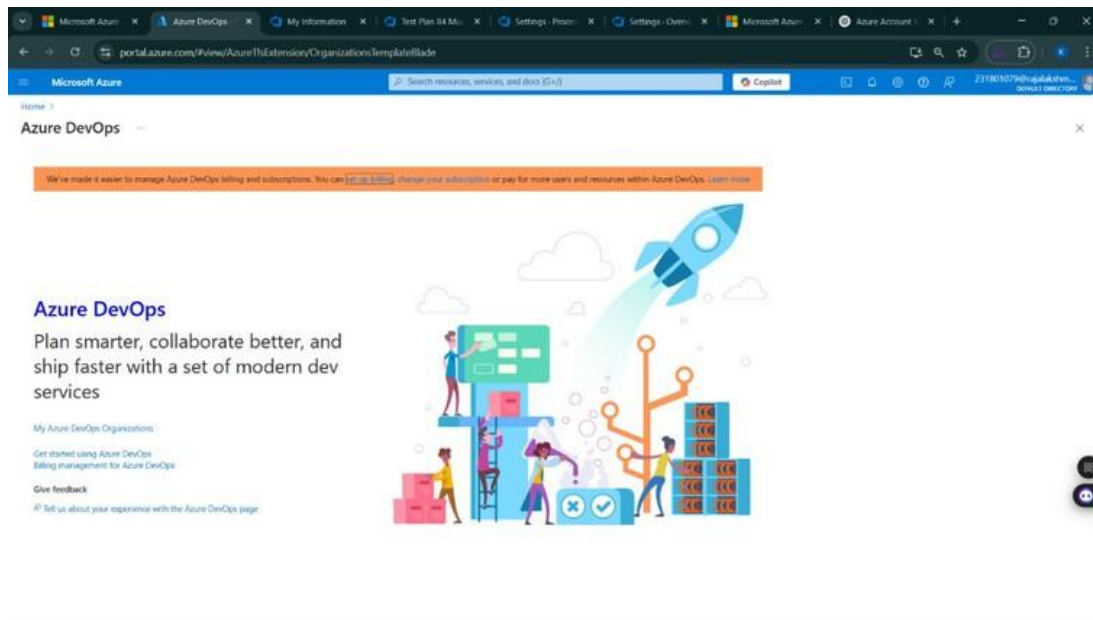
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the *My Azure DevOps Organization* link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

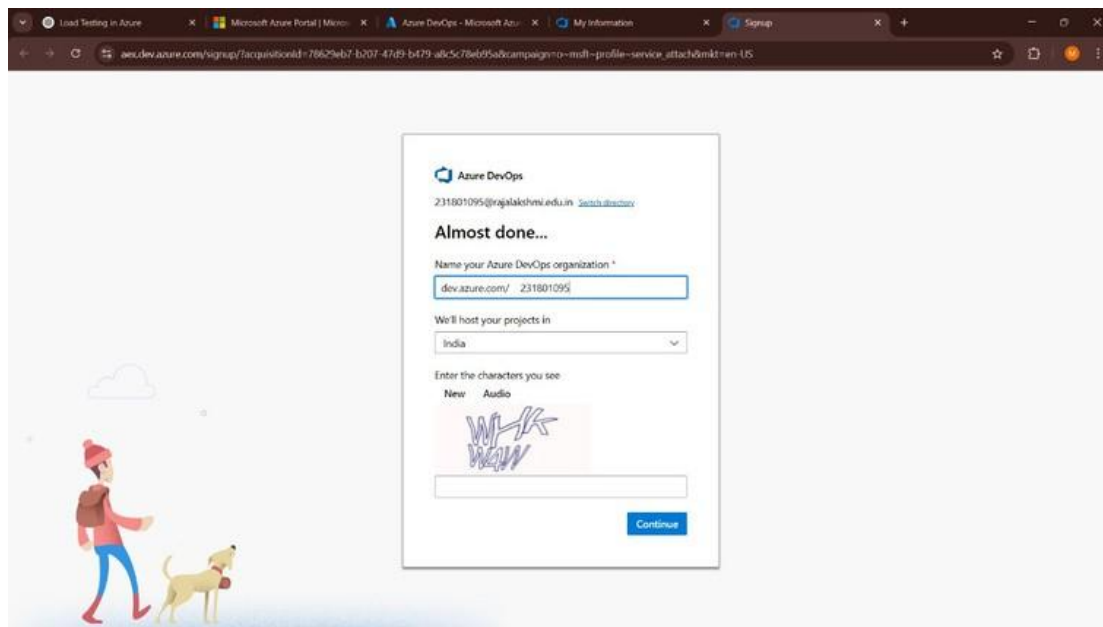
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account



2. Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

dev.azure.com/sivas2024cse/DIGITAL%20CERTIFICATE%20MANAGEMENT%20SYSTEM%20WITH%20BATCH%20GENERATION%20-1

Azure DevOps sivas2024cse / DIGITAL CERTIFICATE MANA... / Overview / Summary

Search

DG DIGITAL CERTIFICATE ... +

Overview

Summary

Dashboards

Wiki

Boards

Repos

Pipelines

Test Plans

Artifacts

Project settings

DG DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENE...

Private Invite

About this project Like 0

A system to generate, manage, and verify digital certificates in batch mode using Azure services.

Project stats Period: Last 7 days

Boards

0 Work items created

0 Work items completed

Repos

0 Pull requests opened

0 Commits by 0 authors

Pipelines

0% Builds succeeded

Members 2

SS AB

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Azure DevOps

sivas2024cse

/

DIGITAL CERTIFICATE MANA...

/

Overview

/

Dashboards

Search

SS

DIGITAL CERTIFICATE ...

+

Overview

Summary

Dashboards

Wiki

Boards

Repos

Pipelines

Test Plans

Artifacts

DIGITAL CERTIFICATE MANAGEMENT S...

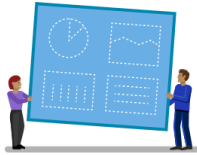
▼

☆

⚙

Modified 7 Feb by siva s

Edit



This dashboard doesn't have widgets just yet!

Add one or more widgets to gain visibility into your team's progress.

Add a widget

4. Project dashboard

STUDENT REG NO

CS23432

5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

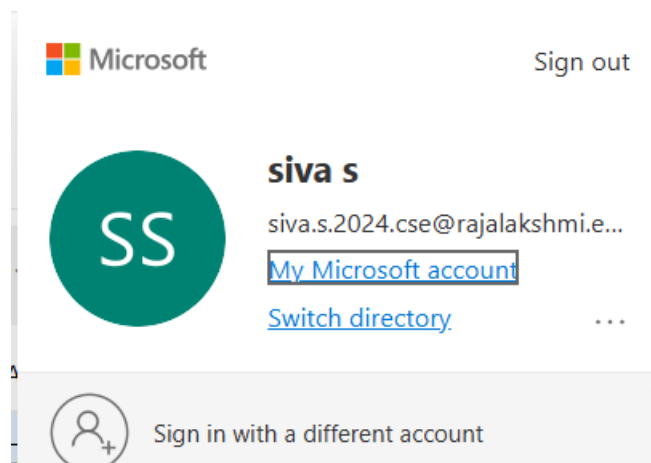
Azure DevOps sivas2024cse / DIGITAL CERTIFICATE MANA... / Boards / Work items

Work items

Recently updated | + New Work Item | Open in Queries | Column Options | Import Work Items | Recycle Bin

Filter by keyword | Types | Assigned to | States | Area | Tags | X

ID	Title	Assigned To	State	Area Path	Tags
28	Print certificate	siva s	Design	DIGITAL CERTIFICATE MANAG...	
27	Download certificate as PDF	siva s	Design	DIGITAL CERTIFICATE MANAG...	
26	Reject certificate successfully	siva s	Design	DIGITAL CERTIFICATE MANAG...	
25	Approve certificate successfully	siva s	Design	DIGITAL CERTIFICATE MANAG...	
24	Verify generated certificate content	siva s	Design	DIGITAL CERTIFICATE MANAG...	
23	Generate certificate for valid student	siva s	Design	DIGITAL CERTIFICATE MANAG...	
22	Create new batch successfully	siva s	Design	DIGITAL CERTIFICATE MANAG...	
21	Verify login with invalid password	siva s	Design	DIGITAL CERTIFICATE MANAG...	
20	Verify login with valid credentials	siva s	Design	DIGITAL CERTIFICATE MANAG...	
14	Certificate Template Design	siva s	To Do	DIGITAL CERTIFICATE MANAG...	
13	Student Data Upload Module	Adithya B	Doing	DIGITAL CERTIFICATE MANAG...	
16	Certificate Verification System	Shabnam Barakkat	To Do	DIGITAL CERTIFICATE MANAG...	
17	Admin Dashboard for Certificate Management	shivam jaiswal	To Do	DIGITAL CERTIFICATE MANAG...	
11	Digital certificate Management with Batch Generation -1	siva s	To Do	DIGITAL CERTIFICATE MANAG...	



Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3	SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING
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Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface. The left sidebar contains navigation options: Overview, Boards, Backlogs (selected), Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area displays the 'Backlog' for the project 'DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION'. It features a table with the following data:

Order	ID	Title	Assigned To	State	Tags
1	17	Admin Dashboard for Certificate Management	shivam jaiswal	To Do	
2	16	Certificate Verification System	Shabnam Barak...	To Do	
3	15	Batch Certificate Generation	shivam jaiswal	To Do	
4	14	Certificate Template Design	siva s	To Do	
5	13	Student Data Upload Module	Adithya B	Doing	

On the right, there is a 'Planning' panel with a 'Sprint 1' summary: '17/02/2026 - 03/03/2026', 'Planned Effort: 24 11 working days', and a count of 5 items. A '+ New Sprint' button is also visible.

1.Fill in Epics

Azure DevOps sivas2024cse / DIGITAL CERTIFICATE MANA... / Boards / Backlogs

ISSUE 14

14 Certificate Template Design

siva s 0 Comments Add Tag

State: To Do Area: DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1 Reason: Moved to backlog Iteration: DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1\Sprint 1 Updated by siva s: 14 Mar

Description

Create customizable certificate templates with placeholders for student details. The system should allow administrators to select templates and preview certificates before batch generation.

Planning

Priority: 2
Effort: 5

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

[Add an existing work item as a parent](#)

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[switch to Markdown editor](#)

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4	SPRINT PLANNING
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Aim:

To assign user story to specific sprint for the Product review and rating system Project.

SprintPlanning Sprint1

The screenshot displays the Azure DevOps Sprints interface for a project named "DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1". The board is set to "Sprint 1" and shows a list of tasks in the "To Do" column. Each task card includes a title, a "To Do" status, and an assigned user. The tasks are:

- 17 Admin Dashboard for Certificate Management** (Assigned to: shivam jaiswal)
- 16 Certificate Verification System** (Assigned to: Shabnam Barakkat)
- 15 Batch Certificate Generation** (Assigned to: shivam jaiswal)
- 14 Certificate Template Design** (Assigned to: siva s)
- 13 Student Data Upload** (Assigned to: [unassigned])

The sprint duration is 17 February - 3 March, totaling 11 work days. The board also includes a "Collapse all" button and a "New Work Item" button.

Sprint 2

DG DIGITAL CERTIFICATE ... +

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1

+ New Work Item

Column Options

Taskboard Backlog Capacity Analytics

Sprint 1 Person: All

17 February - 3 March
11 work days

Collapse all

17 Admin Dashboard Certificate Management

To Do

shivam jaiswal

32 management of reviewed dashboards

To Do

siva s

16 Certificate Verification System

To Do

Shabnam Barakkat

33 evaluation of certificates

To Do

Adithya B

15 Batch Certificate Generation

To Do

shivam jaiswal

14 Certificate Template Design

To Do

To Do

Doing

Done

Sprint 3

Azure DevOps sivas2024cse / DIGITAL CERTIFICATE MANA... / Boards / Sprints

Search

+ New Work Item

Column Options

Taskboard Backlog Capacity Analytics

admin dashboards Person: All

4 March - 18 March
11 work days

Collapse all

34 admin dashboard evaluation

To Do

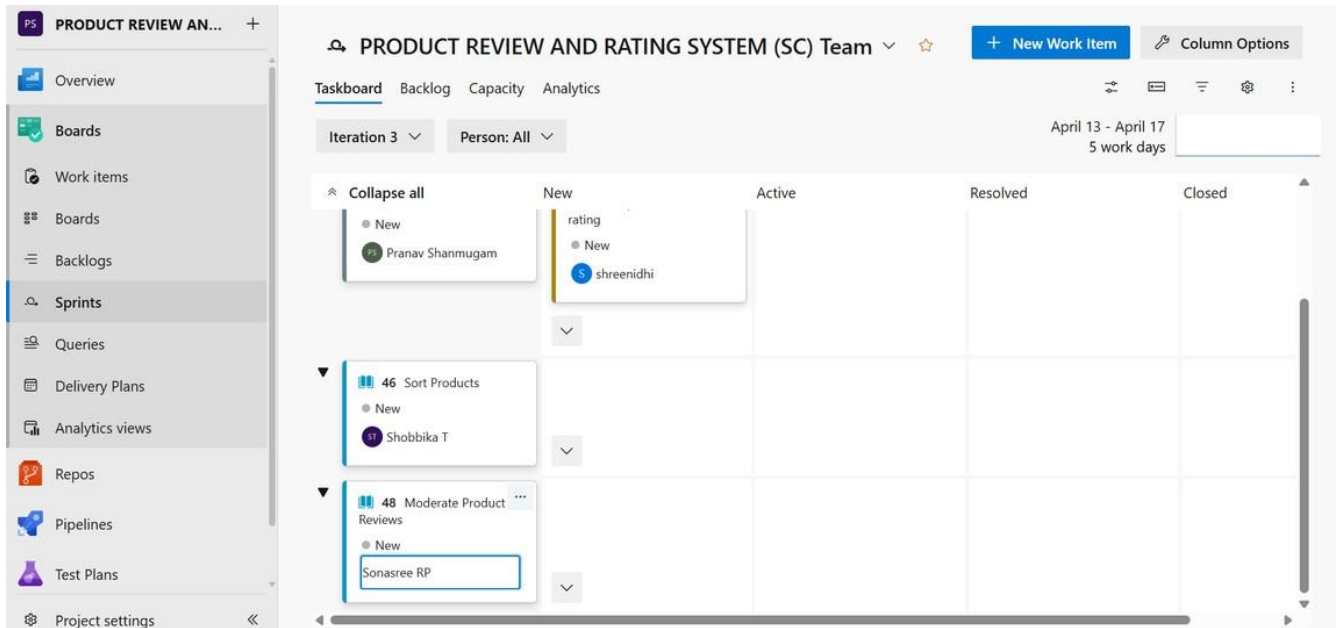
shivam jaiswal

To Do

Doing

Done

Sprint 4



Result:

The Sprints are created for the Product Review and Rating System Project.

EXP NO: 5	POKER ESTIMATION
-----------	------------------

Aim:

Create Poker Estimation for the user stories - Product Review and Rating System.

PokerEstimation

The screenshot displays the Azure DevOps interface for a work item titled "16 Certificate Verification System". The work item is assigned to "Shabnam Barakkat" and is in the "To Do" state. The area is "DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1" and the iteration is "DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1\Sprint 1".

Description: Develop a verification feature where users can validate certificates using a unique certificate ID or QR code to ensure authenticity.

Planning: Priority 2, Effort 3.

Deployment: To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development: Add link. Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work: Add link. [Add an existing work item as a parent](#)

Discussion: Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request. [switch to Markdown editor](#)

Comments:

- siva s commented 14 Mar: hi don't wait for the start up just start working on th project which was assigned to you you are free to use professional outside tools and update the status here

Result:

The Estimation/Story Points is created for the project using Poker Estimation.

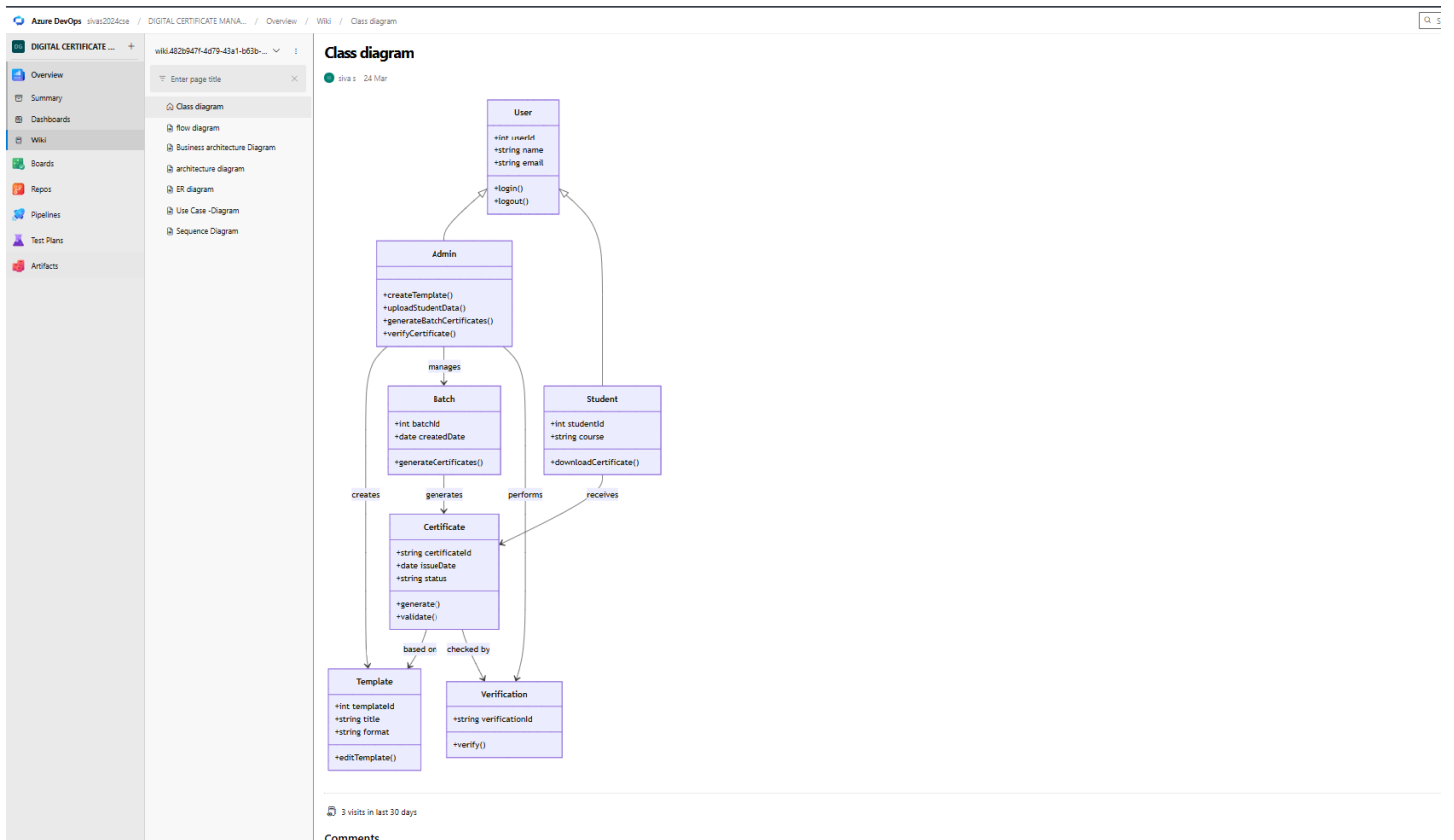
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

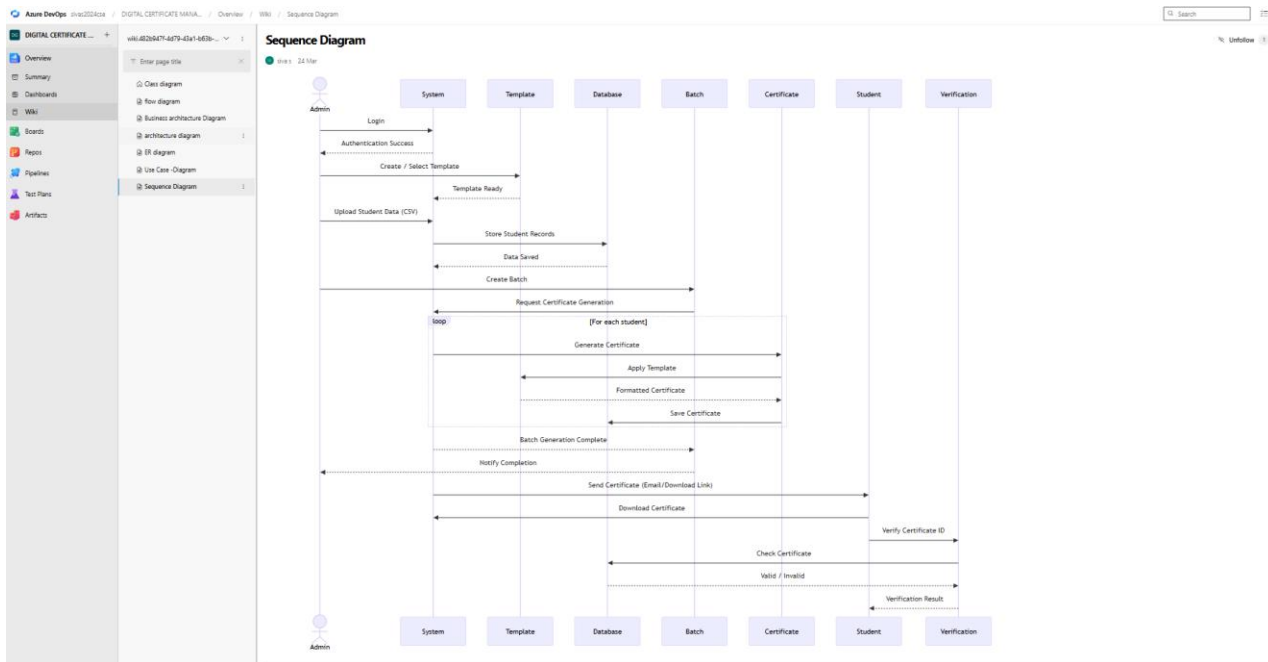
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Product Review and Rating System.

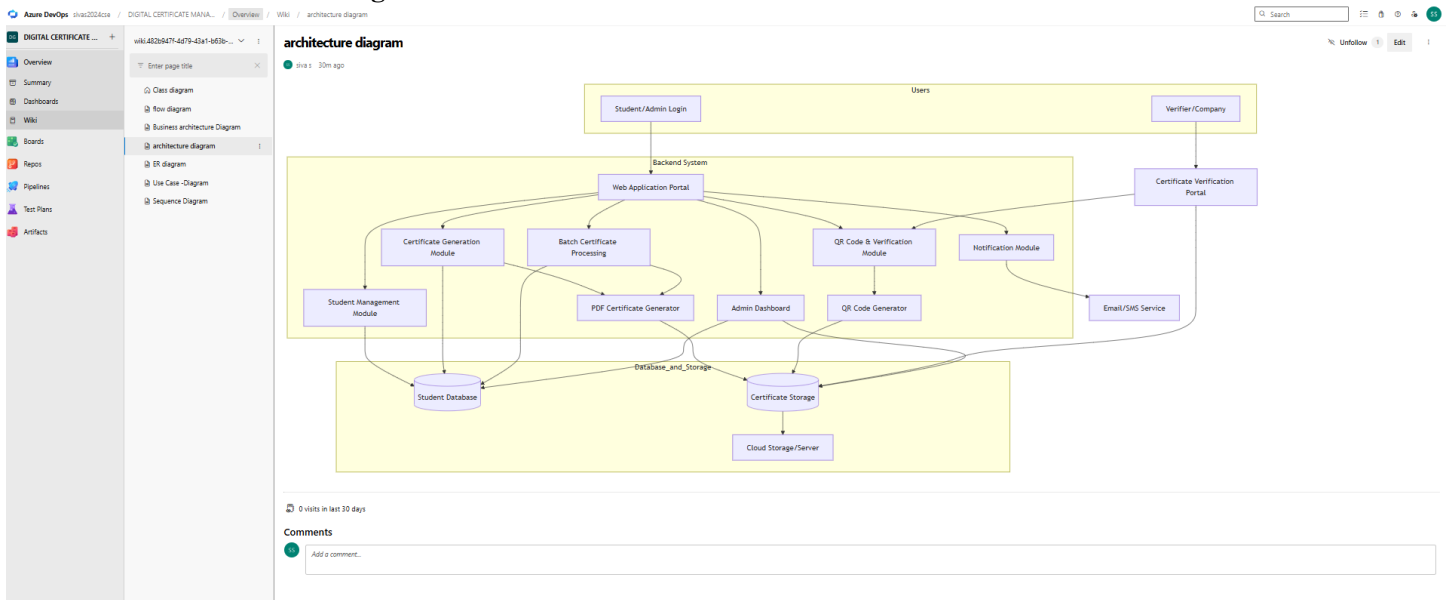
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

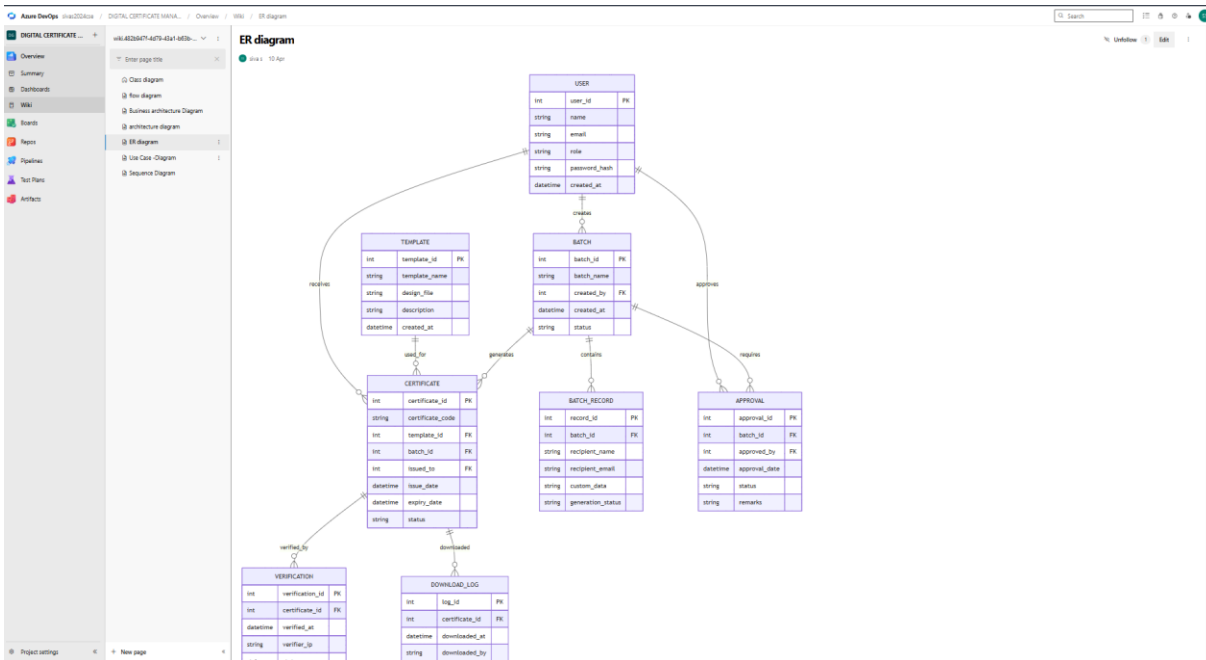
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Product Review and Rating System.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
------------------	--

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Registration & Login
- Viewing and Managing Products
- Adding, Editing, and Deleting Products
- Submitting Reviews and Ratings for Products
- Searching, Filtering, and Sorting Products
- Admin Dashboard for Managing Users and Reviews

2. Define User Interactions

- o Each test case simulates a real user behavior (User logs in successfully, adds a product, submits a 5-star review, or searches for a product successfully.).

3. Design Happy Path Test Cases

- o Focused on validating that all features function as expected under normal conditions.
- o Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- o Simulate negative or unexpected scenarios to test robustness and error handling.
- o Example: Login fails with invalid credentials, review submission fails with empty fields, product deletion fails for unauthorized users, or search returns no products.

5. Break Down Steps and Expected Results

- o Each test case contains step-by-step actions and corresponding expected outcomes.
- o Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- o Test cases are named clearly (e.g., TC01 – Successful User Login, TC02 – Invalid Login Attempt)
- o Helps in quick identification and linking to user stories or features.

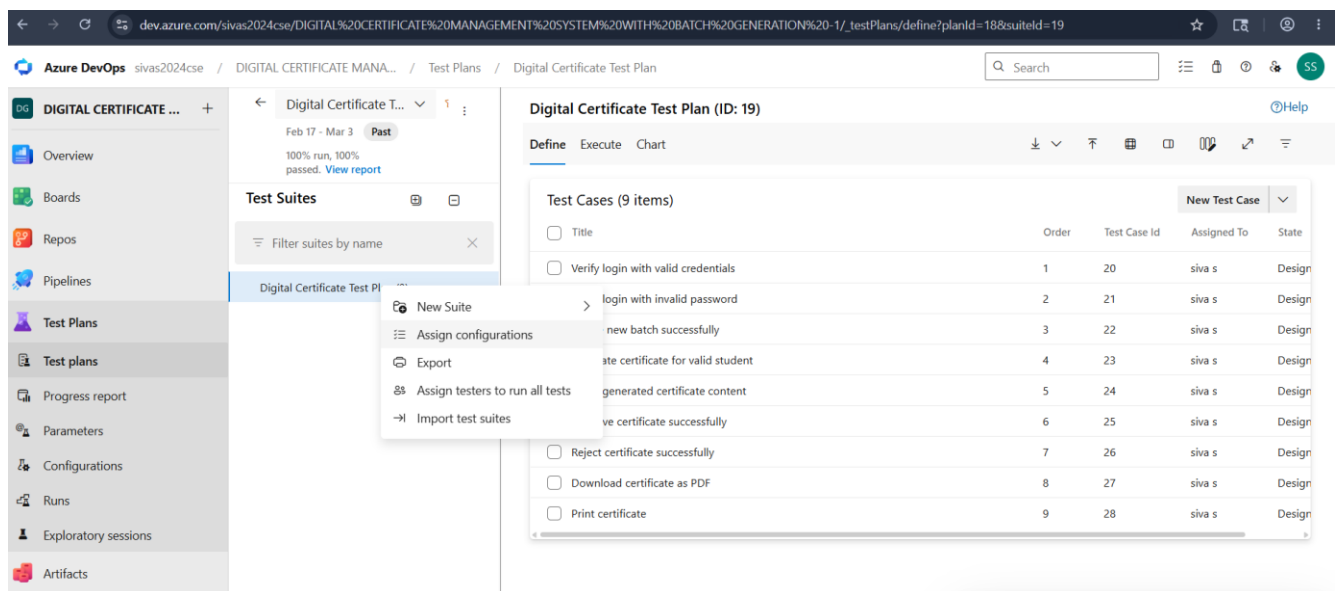
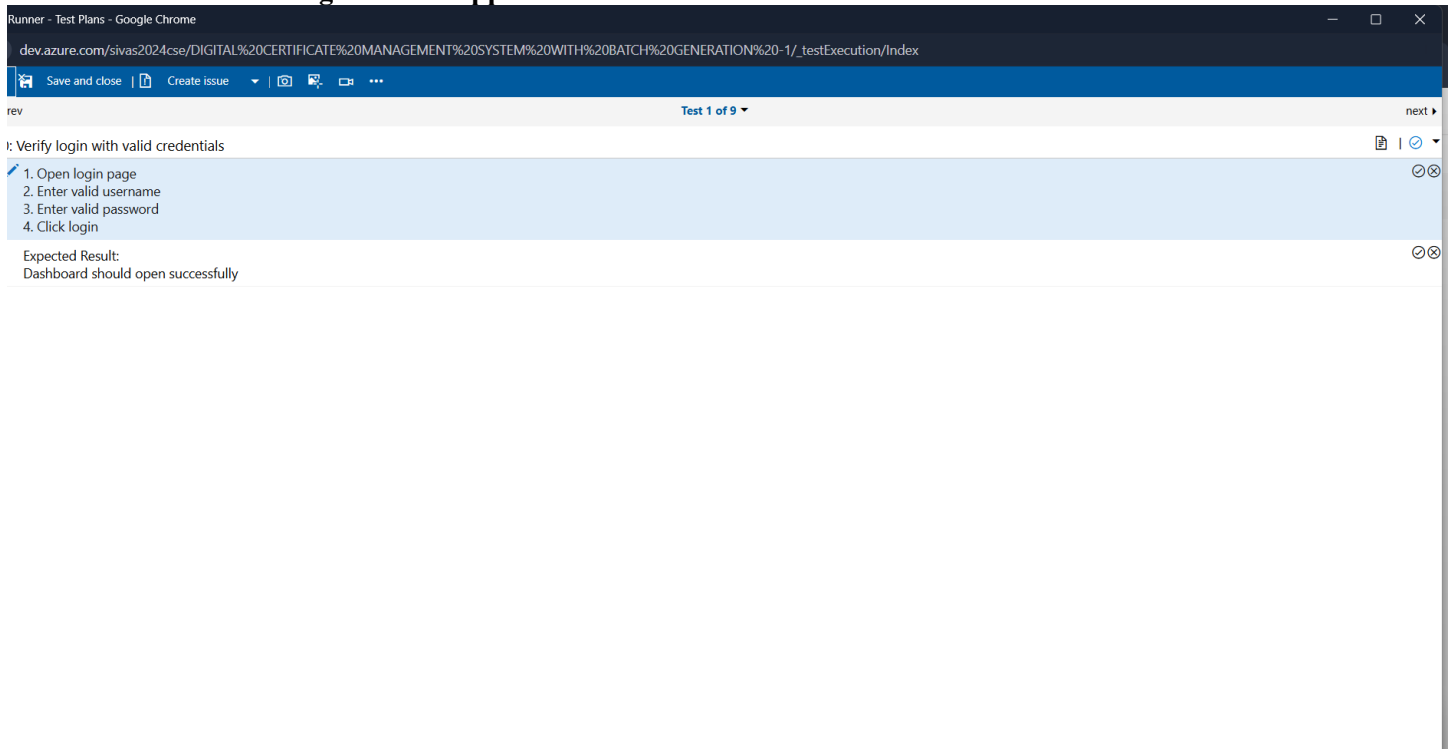
7. Separate Test Suites

- o Grouped test cases based on functionality (e.g., User Management, Product Management, Review & Rating System, Search & Filtering, Admin Features).
- o Improves organization and test execution flow in Azure DevOps.

8. Prioritize and Review

- o Core functionalities like login, product management, and review submission are marked high priority.
- o Reviewed for completeness and traceability against feature requirements.

1 test cases running in online application after test suites creation& test suites creation



1. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Product Review and Rating System – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access the system (ID: 101).
- As a user, I need to view, add, edit and delete products (ID:102)
- As a user, I should be able to submit ratings and reviews for product (ID: 103).
- As a user, I want to search and filter product easily (ID: 104).
- As a admin, I need to manage users, reviews, and products (ID: 105).

Test Suites

Test Suit: TS01 - User Management (ID: 106)

1. TC01 – Successful User Registration

- **Action:**
 - Goto theSign-Up page.
 - Enter valid name, email, and password.
 - Click "Register".
- **Expected Results:**
 - Register form is displayed.
 - Fields accept values without error.
 - Account is created, and the user is redirected to the dashboard.
- **Type:**Happy Path

2. TC02 – Login with Invalid Password

- **Action:**
 - Goto the Login page.
 - Enter valid email and password.
 - Click on "Login".
- **Expected Results:**
 - Login form is displayed.
 - Fields accept data without error.
 - User is logged in andredirected to the dashboard.
- **Type:** Happy Path

3. TC03 – Sign Up with Existing Email

- **Action:**
 - Goto theSign-Up page.
 - Enter a name and an already registered email.

- Click on "Register".
- **Expected Results:**
 - Fields accept data.
 - Error message "Email already registered" is displayed.
- **Type:** Error Path

4. TC04 – Login with Wrong Password

- **Action:**
 - Go to the Login page.
 - Enter valid email and incorrect password.
 - Click on "Login".
- **Expected Results:**
 - Input is accepted.
 - Error message "Invalid username or password" is shown.
- **Type:** Error Path

Test Suit: TS02 - Product Management (ID: 107)

1. TC05 – View Products

- **Action:**
 - Log in successfully.
 - Navigate to "My Products" section.
- **Expected Results:**
 - All created playlists are displayed clearly.
- **Type:** Happy Path

2. TC06 – Products Loading Failure

- **Action:**
 - Disconnect from the internet.
 - Navigate to "My Products".
- **Expected Results:**
 - Network is offline.
 - Error message "Unable to load products" is shown.
- **Type:** Error Path

Test Suit: TS03 - Review and Rating System (ID: 108)

1. TC07 – Submit Review Successfully

- **Action:**
 - Login and Open product details page
 - Enter Review and Click on “Submit Review”
- **Expected Results:**
 - Review added and Rating updated successfully
- **Type:** Happy Path

2. TC08 – Submit Empty Review

- o **Action:**
 - Leave review field blank
 - Click “Submit Review”.
- o **Expected Results:**
 - • Error message "Review cannot be empty" is displayed.
- o **Type:** ErrorPath

Test Suit: TS04 - Search & Filtering (ID: 109)

1. TC09 – Search Product by Name

- o **Action:**
 - Navigate to "My Products".
 - Enter product name in search bar.
 - Click "Search".
- o **Expected Results:**
 - Matching products are displayed successfully.
- o **Type:** HappyPath

2. TC10 – Search for Non-Exsisting Product

- o **Action:**
 - Navigate to “My Products”.
 - Enter invalid or unavailable product name.
 - Click "Search".
- o **Expected Results:**
 - Error message "No Products available" is shown.
- o **Type:** Error Path

3. TC11 – Filter Product by Category

- o **Action:**
 - Navigate to products page.
 - Select a category from filter options.
 - Click "Apply Filter".
- o **Expected Results:**
 - Filter works correctly without errors.
- o **Type:** Happy Path

4. TC12 – Filter Returns No Products

- o **Action:**
 - Navigate to products page.
 - Select a category with
 - No available products.
 - Click "Apply Filter".
- o **Expected Results:**
 - Errormessage: "No products found in this category".
- o **Type:** Error Path

Test Suit: TS05 - Admin Features (ID: 110)

1. TC13 – Admin Deletes Inappropriate Review

- o **Action:**
 - Login as admin.
 - Navigate to review management section.
 - Select a review.
 - Click "Delete".
- o **Expected Results:**
 - Deleted review no longer appears.
- o **Type:** HappyPath

2. TC14 – Unauthorized User Accessing Admin Panel

- o **Action:**
 - Login with valid credentials. Click on
 - "Normal user".
 - Select categories.
 - Attempt to open admin dashboard URL.
- o **Expected Results:**
 - Error message: "Access Denied" or "Unauthorized access".
- o **Type:** Error Path

Test Cases

Azure DevOps sivas2024cse / DIGITAL CERTIFICATE MANA... / Test Plans / Digital Certificate Test Plan

Search

DIGITAL CERTIFICATE ... +

- Overview
- Boards
- Repos
- Pipelines
- Test Plans**
- Test plans
- Progress report
- Parameters
- Configurations
- Runs
- Exploratory sessions
- Artifacts

Digital Certificate Test Plan (ID: 19)

Feb 17 - Mar 3 **Past**
100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Digital Certificate Test Plan (9)

Digital Certificate Test Plan (ID: 19)

Define Execute Chart

Test Cases (9 items)

☐ Title	Order	Test Case Id	Assigned To	State
☐ Verify login with valid credentials	1	20	siva s	Design
☐ Verify login with invalid password	2	21	siva s	Design
☐ Create new batch successfully	3	22	siva s	Design
☐ Generate certificate for valid student	4	23	siva s	Design
☐ Verify generated certificate content	5	24	siva s	Design
☐ Approve certificate successfully	6	25	siva s	Design
☐ Reject certificate successfully	7	26	siva s	Design
☐ Download certificate as PDF	8	27	siva s	Design
☐ Print certificate	9	28	siva s	Design

4. Executing test cases

Azure DevOps sivas2024cse / DIGITAL CERTIFICATE MANA... / Test Plans / Digital Certificate Test Plan

Search

DIGITAL CERTIFICATE ... +

- Overview
- Boards
- Repos
- Pipelines
- Test Plans
- Test plans
- Progress report
- Parameters
- Configurations
- Runs
- Exploratory sessions
- Artifacts

Digital Certificate T...
Feb 17 - Mar 3 **Past**
100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Digital Certificate Test Plan (9)

Digital Certificate Test Plan (ID: 19)

Define **Execute** Chart

Test Points (9 items)

Title	Outcome	Order	Test Case Id	Sta
Verify login with valid credentials	Passed	1	20	De
Verify login with invalid password	Passed	2	21	De
Create new batch successfully	Passed	3	22	De
Generate certificate for valid student	Passed	4	23	De
Verify generated certificate content	Passed	5	24	De
Approve certificate successfully	Passed	6	25	De
Reject certificate successfully	Passed	7	26	De
Download certificate as PDF	Passed	8	27	De
Print certificate	Passed	9	28	De

Run for web application

5.

Test Suite Execution Content

- Created and organized multiple test suites for the Digital Certificate Management System.
- Designed test cases for major modules such as student management, certificate generation, batch processing, and certificate verification.
- Executed all test cases successfully and validated the expected outputs.
- Verified system functionality, input validation, and database operations through manual testing.
- Performed end-to-end testing for certificate generation and QR verification processes.
- Identified and resolved minor issues during test execution to improve system reliability.
- Successfully completed test runs with accurate results and proper workflow validation.
- Ensured that all modules met the required functional and performance expectations.

During the development and testing phase of the Digital Certificate Management System, bugs were identified, created, and managed efficiently using Azure DevOps. Different types of defects related to certificate generation, student data management, batch processing, and QR code verification were detected during system testing. Each bug was documented with detailed information including severity, priority, expected results, actual results, and reproduction steps to ensure proper tracking and resolution. After fixing the issues, retesting was performed to verify that the application functioned correctly without errors. The bug management process helped improve the quality, reliability, and performance of the overall system.

Points

1. Created and tracked bugs systematically during software testing.
2. Documented defects with proper severity and priority levels.
3. Identified issues in certificate generation and verification modules.
4. Verified validation errors in student data processing.
5. Monitored bug lifecycle from active state to resolved state.
6. Performed retesting after bug fixes were completed.
7. Ensured stable system performance after resolving defects.
8. Improved application reliability through effective bug management.

6. Creating the bug

dev.azure.com/shobbik2024cse/PRODUCT%20REVIEW%20AND%20RATING%20SYSTEM%20(SC)/_testExecution/Index

Save and close Create bug

71*: add review

1. Navigate to product page (Precondition: User is logged in)

COMMENT

Add notes or context

2. Enter review text

COMMENT

Add notes or context

3. Click Submit

EXPECTED RESULT

Review is successfully added and displayed

Runner - Test Plans - Google Chrome

dev.azure.com/shobbik2024cse/PRODUCT%20REVIEW%20AND%20RATING%20SYSTEM%20(SC)/_testExecution/Index

NEW BUG *

Unable to submit review. Takes more time when there's poor network

No one selected 0 Comments Add Tag Save and Close

State: New Area: PRODUCT REVIEW AND RATING SYSTEM (SC)

Reason: New Iteration: PRODUCT REVIEW AND RATING SYSTEM (SC)/Iteration 1

Details 2 0

5/11/2026 2:22 PM Bug filed on "add review"

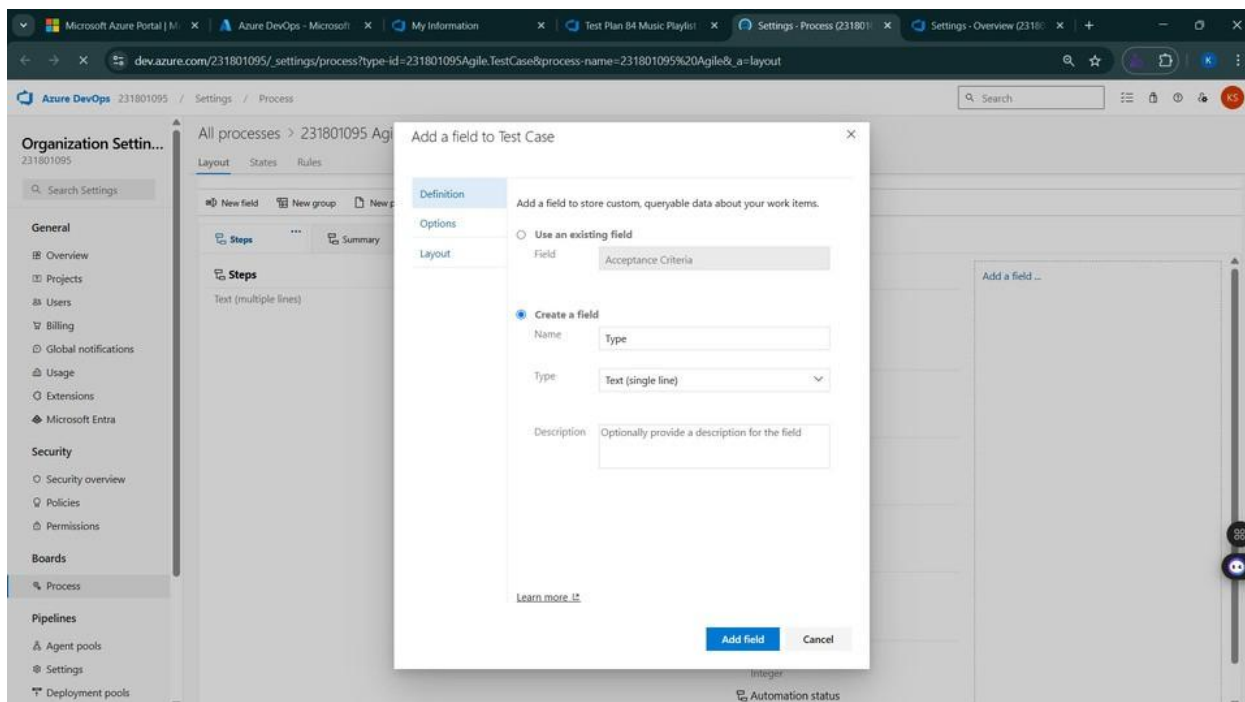
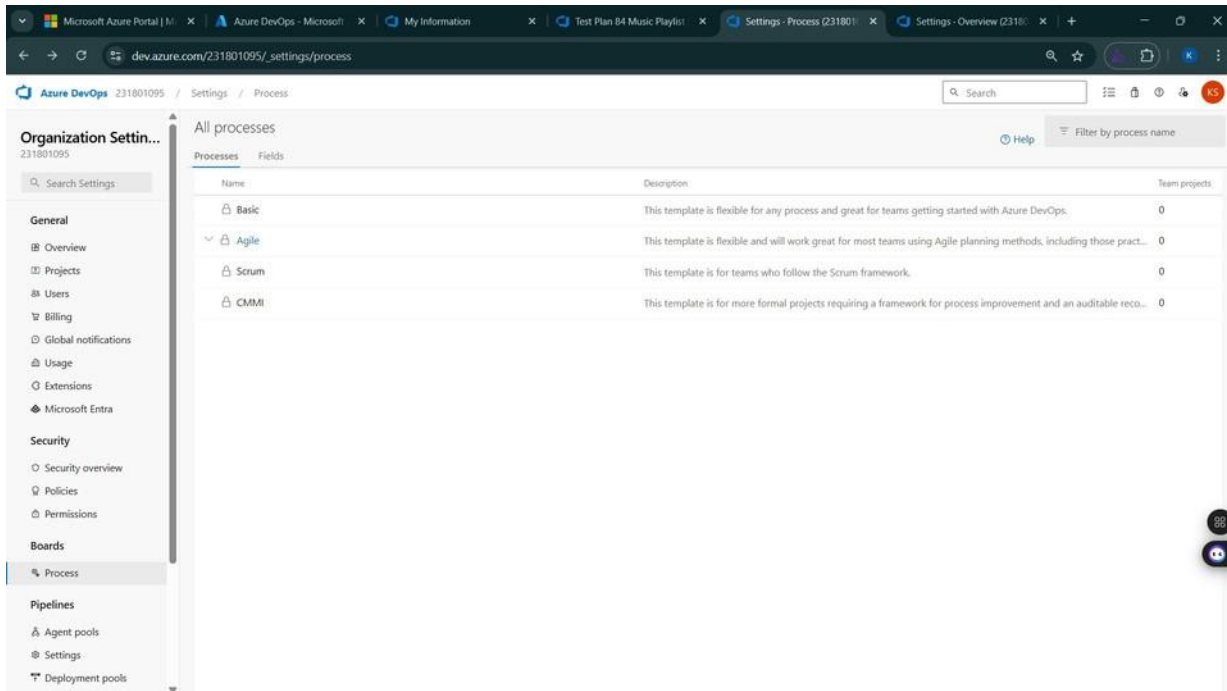
Step no.	Result	Title	Story Points
1.	Passed	Navigate to product page (Precondition: User is logged in)	Priority 2
2.	Passed	Enter review text	Severity 3 - Medium
3.	Failed	Click Submit	Activity

Expected Result

Results:

In azure devops test cases and test suites and bug detection and successful test cases passing is done and implemented successfully

7 changing the templete



Microsoft Azure Portal | Microsoft Azure DevOps - Microsoft | My Information | Test Plan 84 Music Playlist | Settings - Process (231801095) | Settings - Overview (231801095)

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

Filter by process name

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

Microsoft Azure Portal | Microsoft Azure DevOps - Microsoft | My Information | Test Plan 84 Music Playlist | Settings - Process (231801095) | Settings - Overview (231801095)

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

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- Process

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- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

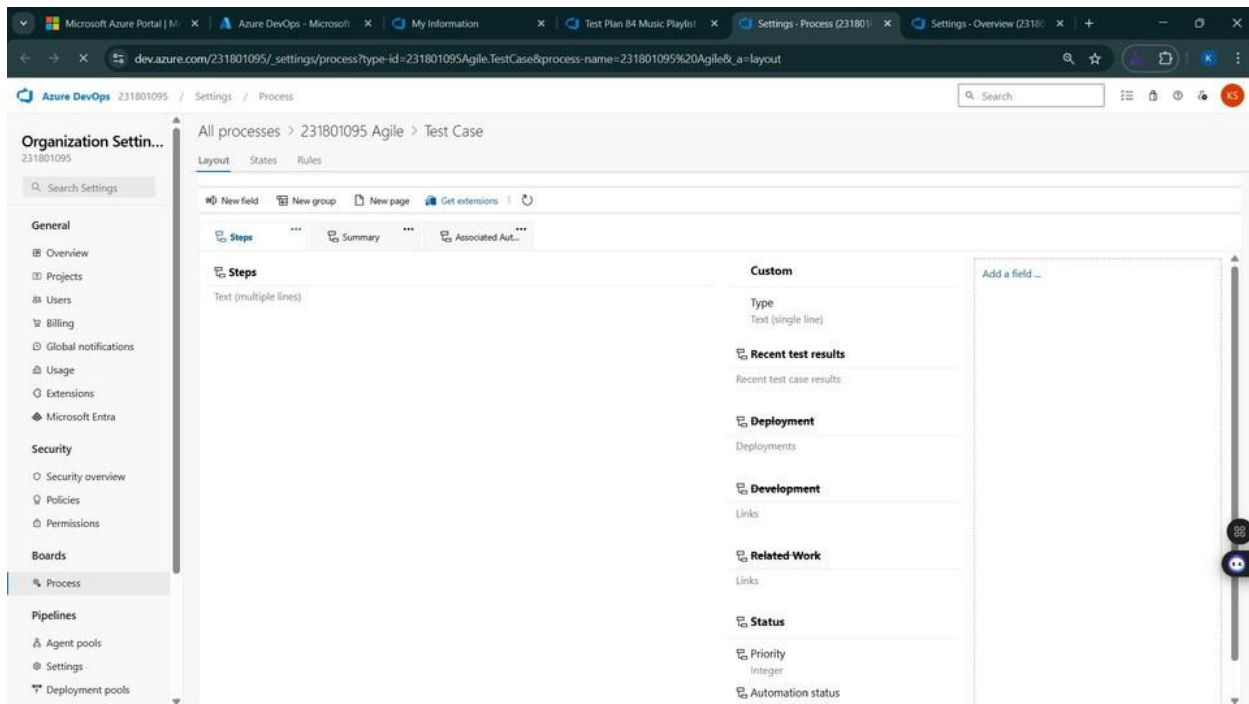
Filter by process name

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
231801095 Agile (default)		1
Agile Plus		0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

8)View the new test case template

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path



EXP NO: 9	LOAD TESTING AND PIPELINES
------------------	-----------------------------------

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

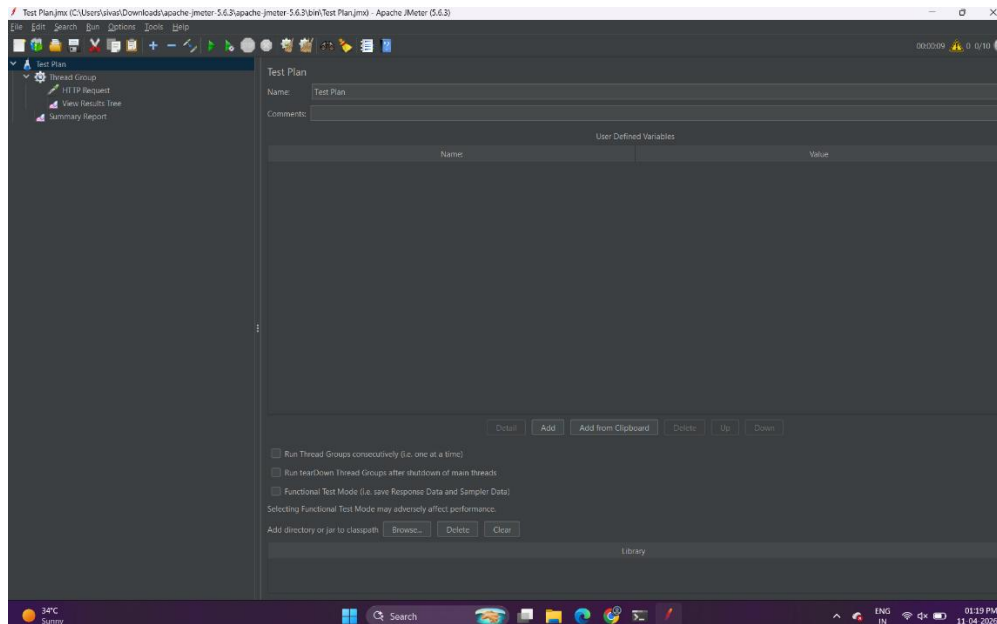
Steps to Create and Run a Load Test:

Once your resource is ready:

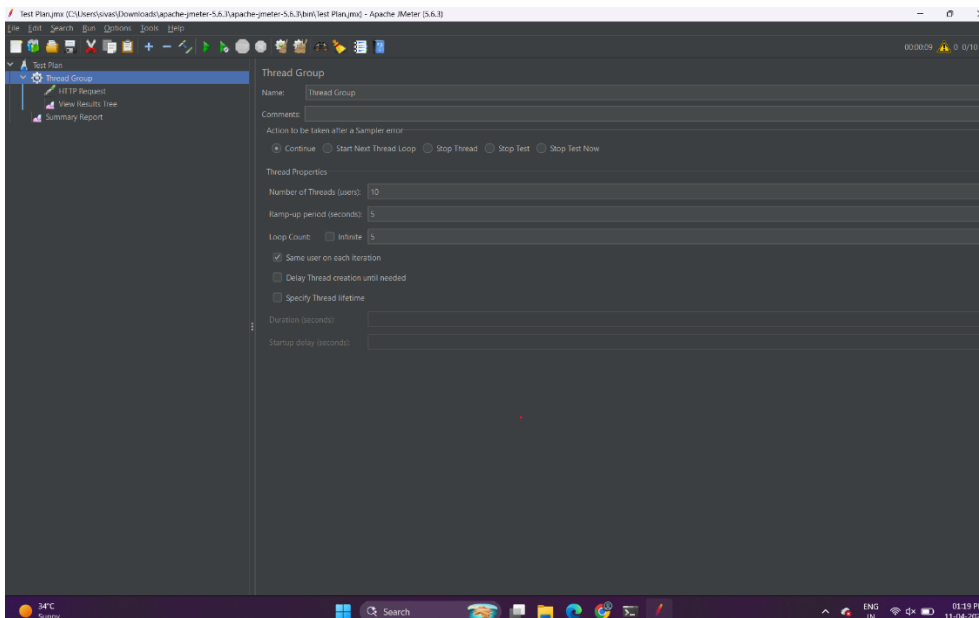
1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o *Test Name*: Provide a unique name.
 - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings

This load testing was done using external application jmeter where we need to extract the zip file and download json to support the jmeter using this jmeter objects were created and progress report was done.

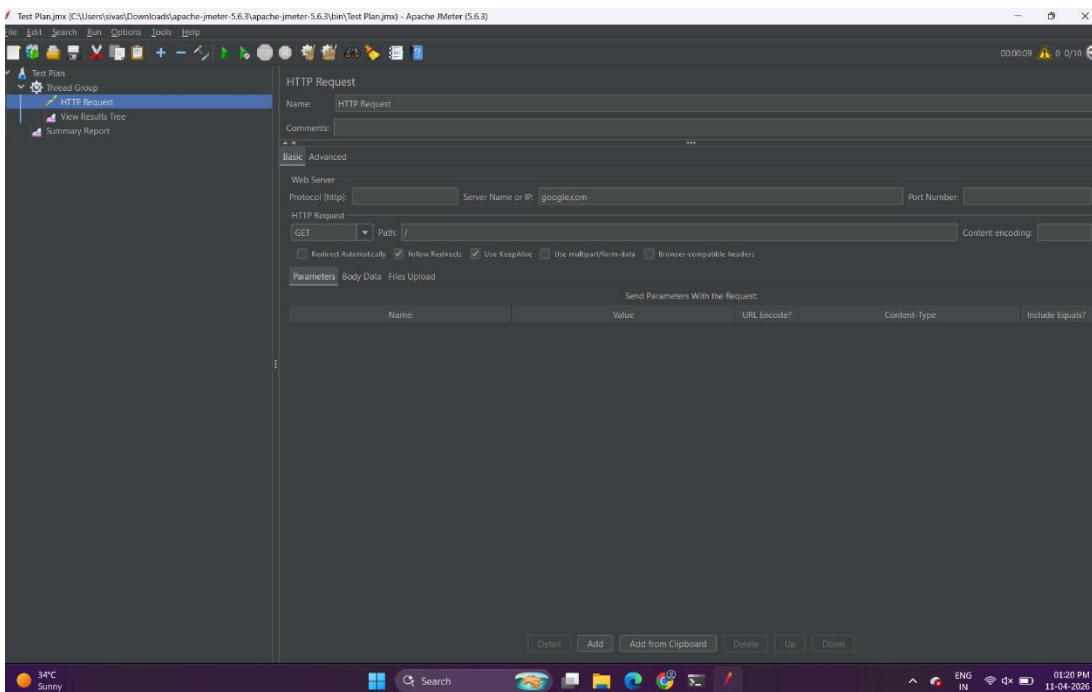
1) Load testing creating objects



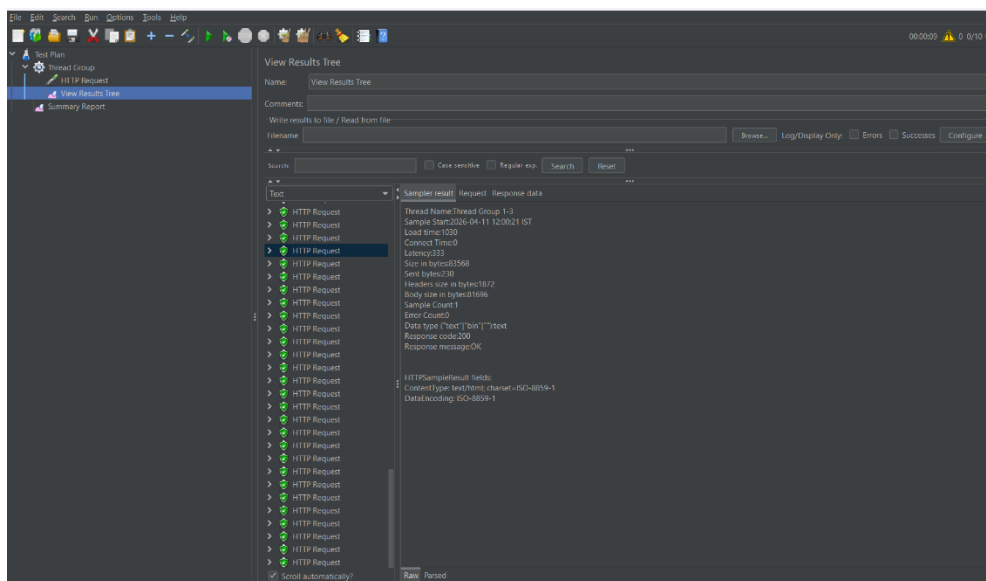
2) creation of threads



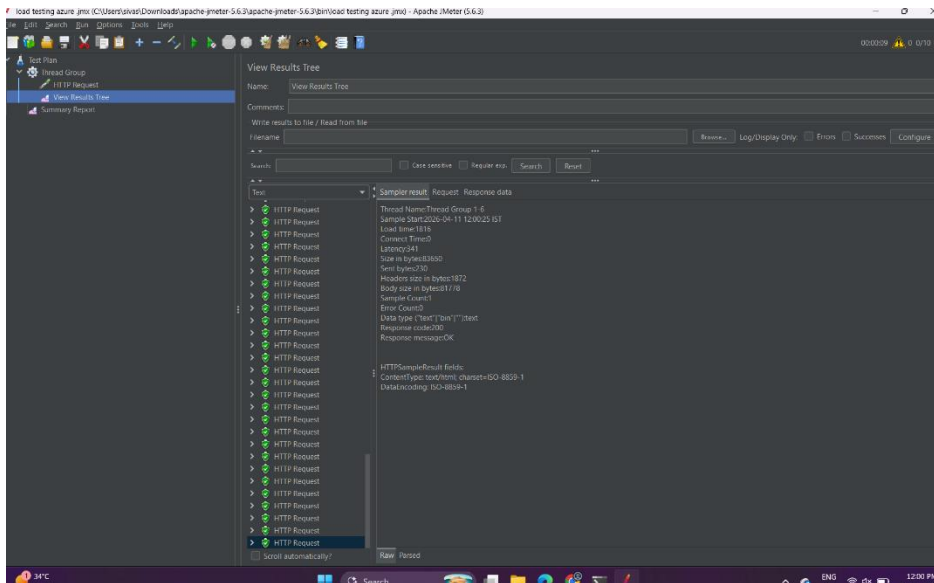
3)setting up the path



4)server request acceptance



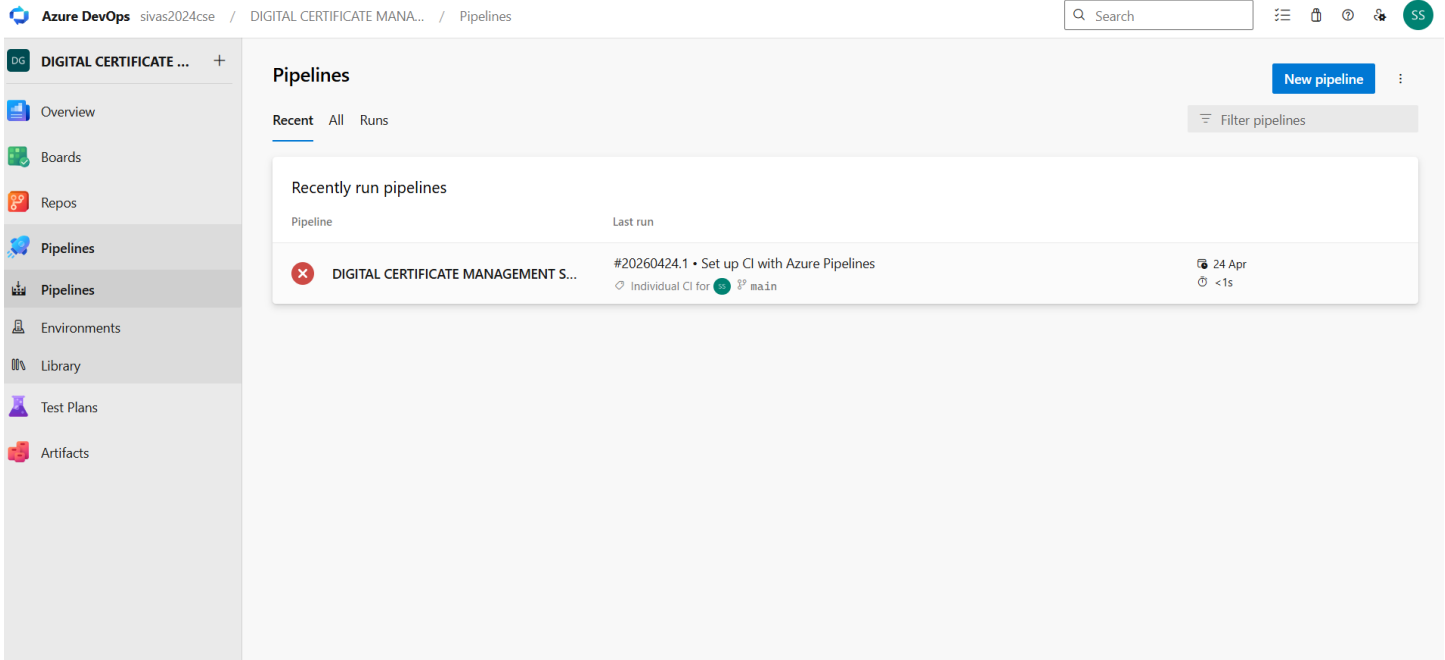
5)Progress acceptance http request



6) overall summary report

The screenshot shows the Apache JMeter 5.6.3 interface with the 'Summary Report' selected. The report displays a table with performance metrics for 10 samples.

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	103	2861	875	28615	4112.69	0.00%	7.6/sec	10.36	0.03	81731.5
TOTAL	103	2861	875	28615	4112.69	0.00%	7.6/sec	10.36	0.03	81731.5



Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - o In Azure DevOps, create a new project.
 - o Create a pipeline and select GitHub as the source.
 - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - o Add the following basic pipeline configuration for Python and Flask:

yml Code

trigger:

-main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

#Step 1: Checkout the code from GitHub

-checkout: self

#Step 2: Set up Python environment

-task: UsePythonVersion@0

inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

#Step 3: Install dependencies from the correct path

-script: |

python -m pip install --upgrade pip

pip install -r project/requirements.txt # Adjusted path to requirements.txt

displayName: "Install dependencies"

#Step 4: Run a simple Python script to check the environment

-script: |

python -c "print('Rate and review your favourite products.')"

displayName: "Run a Python script"

3. Pipeline Tasks Include:

- o Setting up the Python environment using the UsePythonVersion task.
- o Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- o Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- o Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- o Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Running the jobs in pipelines

The screenshot displays the Azure DevOps web interface. The top navigation bar shows the user 'sivas2024cse' and the current path: 'DIGITAL CERTIFICATE MANA...' / 'Pipelines' / 'DIGITAL CERTIFICATE MANA...' / '20260515.1'. A search bar is located on the right. The left sidebar contains a list of navigation items: Overview, Boards, Repos, Pipelines (selected), Pipelines (sub-item), Environments, Library, Test Plans, and Artifacts. The main content area is titled 'Jobs in run #20260515.1' and shows a list of jobs. The 'Job' tab is selected, displaying a log of the job's execution. The log shows the following steps:

- 1 Pool: Azure Pipelines
- 2 Image: ubuntu-latest
- 3 Queued: Just now [manage_parallel_jobs]
- 4
- 5 This agent request is not running because you have reached the maximum number of requests that can run for parallelism type
- 6 • Job preparation parameters

Acceptance error :

Due to Microsoft account needs an real account for organisation this jobs was not runned successfully because Microsoft does not allow parallel execution method in creation of pipelines jobs which was the primary goal of this experiment. As paid version we can execute our successfully . but we can run manually to verify our jobs and creation of pipelines

Pipeline manually running

Azure DevOps / DIGITAL CERTIFICATE MANA... / Repos / Commits / DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1

Search

DIGITAL CERTIFICATE ...

- Overview
- Boards
- Repos
- Files
- Commits
- Pushes
- Branches
- Tags
- Pull requests
- Advanced Security
- Pipelines
- Test Plans
- Artifacts

Set up CI with Azure Pipelines...

b526b2bd siva s committed 24 Apr 8:20 main succeeded

Files Details

Parent 1 → This commit Filter 1 changed file Inline

DIGITAL CERTIFICATE MAN...

azure-pipelines.yml

```
1 trigger:
2   - main
3
4 pool:
5   vmImage: 'ubuntu-latest'
6
7 steps:
8   - script: echo "Building Digital Certificate Management System"
9     displayName: 'Build Project'
10
11   - script: echo "Pipeline executed successfully"
12     displayName: 'Success'
```

Azure DevOps sivas2024cse / DIGITAL CERTIFICATE MANA... / Pipelines / DIGITAL CERTIFICATE MANA... / 20260515.1

Search

DIGITAL CERTIFICATE ...

- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Library
- Test Plans
- Artifacts

#20260515.1 • Set up CI with Azure Pipelines

DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BATCH GENERATION -1

Run new

This run is being retained as one of 3 recent runs by main (Branch). View retention leases

Summary Code Coverage

Manually run by siva s

Today at 21:26 7s

Repository and version

DIGITAL CERTIFICATE MANAGEMENT SYSTEM WITH BAT... 0 work items

8:20 main b526b2bd 0 published

Tests and coverage

Get started

View 2 changes Parameters

Stages Jobs

Stage

1 job completed 5s

Job 2s

Rerun Stage

Thereby our pipeline was created and successfully runned manually by the user to the required stages were obtained this alterier method was easily implemented successfully.

Result:

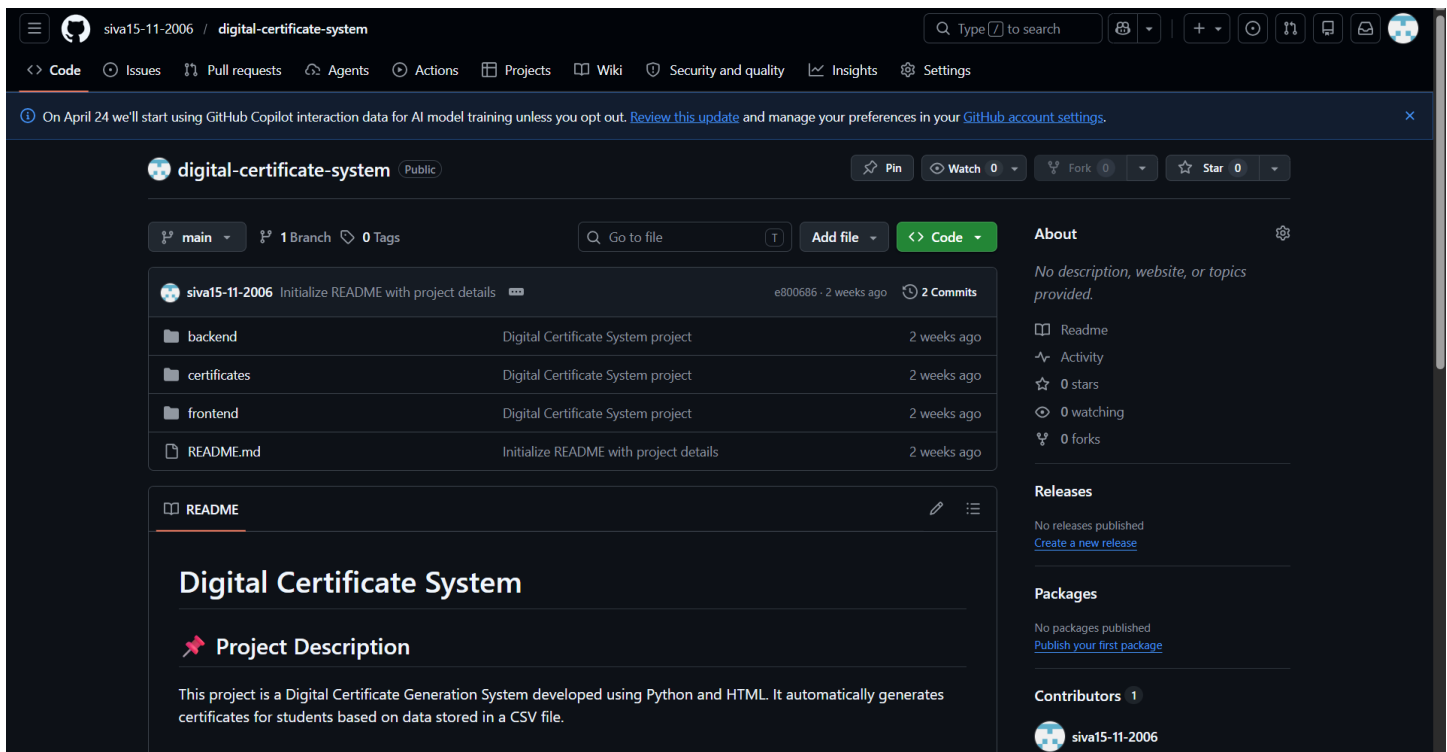
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

EXP NO: 10	GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS
------------	---

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Product review and rating system project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.

